A positive learning window above the critical tanh prefactor

ChatGPT-6 Astra Kaarel Hänni Dmitry Vaintrob*

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Abstract

We extend a critical Boolean tanh posterior upper bound to prefactors $t = 1 + \delta$ just above one, retaining the original Gaussian width powers, global output-margin conditioning, and exact labels. If $S = 1 + \log((d + H + 2)A_{\mathrm{F}}(f))$ and $H \geq CS$, then every realizable width has sufficient sample complexity $\exp(O(S + \delta H))$, times polynomial statistical factors. Thus polynomial Fourier variation, including every parity, retains polynomial sample complexity when $\delta H = O(\log(d + H))$. The proof combines a scalar reference-change bootstrap with an actual finite-width target-mass construction whose perpendicular Gaussian covariance is never replaced. The critical scalar and finite-width compression theorems are stated explicitly as imported inputs. This positive window lies above the canonical range; it does not resolve the unrestricted Boolean depth question at $t \leq 1$.

1 Model and conclusion

For $x \in \{-1, 1\}^d$ use the augmented input $(1, x)$, $H \geq 1$ hidden layers of width N with activation tanh, no internal biases, and a scalar linear readout $z_W(x)$. The independent Gaussian weight variances are t^2/d in the first layer, t^2/N in later hidden layers, and t^2/N^2 in the readout. Write $\mu_{N,t}$ for this unconditioned prior and

$$E_f = \{W : f(x)z_W(x) > 1/2 \text{ for every } x\}, \qquad E_{\mathrm{mar}} = \{W : |z_W(x)| > 1/2 \text{ for every } x\}.$$

The posterior is $\mu_{N,t}$ conditioned first on E_{mar} and then on the exact labels of the training sample. Training inputs are drawn independently from the same distribution used to measure test error. At a test input it fails when the posterior probability of the correct label is at most $2/3$. A width is *realizable* here if $\mu_{N,t}(E_f) > 0$.

Use the uniform Fourier expansion $f(x) = \sum_{T \subseteq [d]} \widehat{f}(T) \prod_{i \in T} x_i$ and define

$$A_{\mathrm{F}}(f) = \sum_T |\widehat{f}(T)|, \quad S = 1 + \log((d + H + 2)A_{\mathrm{F}}(f)), \quad U = S + \delta H.$$

Since f is Boolean, $A_{\mathrm{F}}(f) \geq 1$.

Theorem 1. *There are universal $\delta_0, C, c > 0$ such that the following holds. Let $t = 1 + \delta$, $0 \leq \delta \leq \delta_0$, and $H \geq CS$. At every finite realizable width, for every test distribution on the cube and $0 < \epsilon, \beta < 1/2$, it suffices to take*

$$n \geq \frac{C}{\epsilon}\left\{e^{CU}(1 + \log(1/\epsilon))^c + \log(1/\beta)\right\} \tag{1}$$

to make the confidence-failure set have test mass at most ϵ with training probability at least $1 - \beta$.

*Authors are listed in alphabetical order.

For $A_{\mathrm{F}}(f) \le (d+H)^a$ with fixed a, the bound is polynomial in $d, H, \epsilon^{-1}, \log(1/\beta)$ whenever $\delta H = O(\log(d+H))$. In particular $A_{\mathrm{F}}(f) = 1$ for every parity, regardless of its degree. A fixed positive δ is not covered by a polynomial-depth conclusion. No bound on Fourier variation is inferred from an arbitrary bounded-weight deep tanh teacher.

The proof extends the critical entropy bound to $t > 1$ by changing its Gaussian reference. A target-channel construction then lower-bounds the actual finite-width target mass. The intrinsic transfer theorem combines these estimates, and a VC argument covers the smaller widths. The scalar and intrinsic theorems are stated as imported inputs below; the changes for $t > 1$ are proved here.

2 Scalar comparison and an intrinsic bound

The critical scalar value $V(q)$ is the infimum, over finite scalar paths starting at variance q, of

$$\sum_j \mathrm{KL}(\eta_j \| \mathcal{N}(0, q_{j-1})) + \frac{1}{8q_m}, \qquad q_j = \mathbb{E}_{\eta_j} \tanh^2 G.$$

Write $V_* = \lim_{q \downarrow 0} V(q)$. This is the unrestricted critical scalar baseline; no Gaussian restriction on η_j is imposed.

We use the following critical scalar theorem from the existing critical analysis [1].

Imported theorem 1 (Critical occupation)**.** *For $q_0 \in [1,2]$, suppose $e_j \ge 0$ dominate the actual critical scalar transition entropies, and $r \ge 1/(8q_H)$. If $\sum_{j=1}^H e_j + r \le V_* + a$, $a \ge 0$, then*

$$\sum_{j=1}^H \sqrt{e_j} \le C \log(H+2) + C\sqrt{Ha}.$$

The statement permits upper costs rather than exact marginal entropies: scalar dissipation and Bellman suffix lower bounds remain valid with those larger costs. The same analysis gives $V(q) \le V_*$ and increasing paths from small p with total cost at most $V(p)+p$, length at most C/p, and square-root entropy sum at most $C\log(2/p)$.

A controlled path on a finite set of inputs has full row entropies D_j relative to $\mathcal{N}(0, t^2 Q_{j-1})$. Its output vector must satisfy $z \in \operatorname{range} Q_H$; its readout action is then $r_t = z^T Q_H^\dagger z/(2t^2)$, where $\dagger$ denotes the Moore–Penrose inverse. Outputs outside this range have infinite action. Its total action is $A = \sum D_j + r_t$. Assume one witness input has $|z(x_0)| \ge 1/2$ and initial variance in $[1,2]$. Let $R(D) \ge 1$ solve $R - 1 - \log R = 2D$.

Lemma 1 (Above-critical occupation)**.** *For $0 \le \delta \le \delta_0$, if $A \le V_* + a$, $a \ge 0$, then*

$$Z := \sum_j \sqrt{D_j} \le C\{\log(H+2) + \sqrt{Ha} + \delta H\}, \tag{2}$$

$$\log \prod_j t\sqrt{R(D_j)} \le C\{\log(H+2) + \sqrt{Ha} + \delta H\}. \tag{3}$$

Proof. At x_0 write $p = q_{j-1}$, $d_j = \mathrm{KL}(\eta_j \| \mathcal{N}(0, t^2 p)) \le D_j$, and $M_j = \mathbb{E}_{\eta_j} G^2/p$. All these variances are positive, since a zero variance cannot revive at finite entropy. Put $b = 1 - t^{-2}$ and $c_t = (t^{-2} - 1 + 2\log t)/2$. Gaussian density comparison gives the exact identity

$$d_{1,j} := \mathrm{KL}(\eta_j \| \mathcal{N}(0,p)) = d_j - c_t + \frac{b}{2}(M_j - 1). \tag{4}$$

Entropy duality gives $M_j \le t^2 R(D_j)$ and $R(D) - 1 \le C(\sqrt{D} + D)$. Hence

$$d_{1,j} \le (1 + C\delta)D_j + C\delta\sqrt{D_j} + C\delta^2.$$

For a sufficiently large fixed C_0, take $e_j = (1 + C_0\delta)D_j + C_0\delta\sqrt{D_j} + C_0\delta^2$. These nonnegative costs dominate both D_j and $d_{1,j}$. The critical terminal cost $r_1 = t^2 r_t$ is at least $1/(8q_H)$, and

$$\sum e_j + r_1 \le V_* + C\{a + \delta(1 + a) + \delta Z + H\delta^2\}.$$

Imported Theorem 1 implies

$$Z \le C\log(H + 2) + C\sqrt{Ha + H\delta(1 + a) + H\delta Z + H^2\delta^2}.$$

Put $Y = \log(H + 2) + \sqrt{Ha} + H\delta$. The terms independent of Z under the root are at most CY^2. Thus $Z \le CY + C\sqrt{H\delta Z}$, and Young's inequality absorbs the last term into $Z/2 + CH\delta$. Finally $\log R(D) \le 2\sqrt{D}$ and $\log t \le \delta$ give the gain assertion. □

For completeness we state precisely the finite-width input used with this lemma. Set $a_{N,t}(f) = -N^{-1}\log \mu_{N,t}(E_f)$.

Imported theorem 2 (Intrinsic transfer)**.** *The finite-witness dual certificate and normalized importance compression of [2, 3], with Lemma 1 replacing their critical gain estimate, give sufficient sample size*

$$\frac{C}{\epsilon}\left\{[(d + H + t^2 a_{N,t}(f) + 2)(1 + \log(1/\epsilon))]^c \exp\left(C\left[\sqrt{H(B - V_*)_+} + \delta H\right]\right) + \log(1/\beta)\right\}. \quad (5)$$

Here $B = a_{N,t}(f) + 4/(Ht^2)$. *This holds at every actual finite realizable width, uniformly for* $1 \le t \le 1 + \delta_0$.

Here the population compression depends on the exact gains $t\sqrt{R(D_j)}$; its dependence on t is bounded on this compact interval. The finite-event certificate allows arbitrary Gaussian reference covariances. Its covariance-transfer tolerance has inverse logarithm $O(H\log(2 + t^2 A) + \log(H + m + 2))$, using the coarse gain $\sqrt{C(1 + t^2 A)}$. The readout repair costs $4/(Ht^2) = O(1/H)$. Thus the only sign-sensitive input replaced above is the scalar gain bound; no covariance change in all weight coordinates is used. The certificate divides by the actual raw target mass. Small widths are completed by the ordinary tanh parameter-count VC bound, and the inclusive confidence threshold follows from posterior Gibbs error.

3 An actual finite-width target witness

Proposition 1. *Under the hypotheses of Theorem 1, at every integer* $N \ge \exp(CU)$,

$$\mu_{N,t}(E_f) \ge \exp\{-N[V_* + CU^2/H]\}. \quad (6)$$

We give the complete modifications of the critical scalar-channel construction [1, 4]. Its Fourier seed and bounded-score finite-channel concentration lemma are imported; their dependence on the prefactor is checked below.

Seed and cooling. The Fourier mixture seed, retaining its rare-component tag under both controlled and reference laws, has

$$\mathbb{E}b_1^2 = 1, \quad \mathbb{E}[b_1 h_1(x)] = \sqrt{v_1} f(x), \quad D_1 \le H^{-1}, \quad v_1 \ge e^{-CS}, \quad |b_1| \le e^{CS},$$

and constant feature diagonal. Use the actual first-layer covariance $t^2 I/d$. Each prescribed mean shift has no larger entropy than at $t = 1$, and its noise variance stays in a fixed compact interval. The saturated threshold detectors, cube-translation parity projection, and Fourier mixture therefore retain the same bounds uniformly in t.

Let $F(q) = \mathbb{E}\tanh^2(\sqrt{q}Z)$. The critical drift and the bound $0 \le F'(q) \le 1$ imply, for a large fixed K,

$$q - F(t^2 q) \ge cq^2 \quad (K\delta \le q \le 1). \tag{7}$$

Choose $\Delta \ge K\delta$, small. Free layers reach $q \le \Delta$ in $O(1/\Delta)$ steps; the first such value lies in $[\Delta/2, \Delta]$. For a component $vff^T \preceq Q$ the next first-Hermite component is $t^2 v[1 - F(t^2 q)]^2$. Its ratio to the new diagonal, divided by v/q, is $1 + O(q^2)$ near zero and has a uniform positive lower bound elsewhere. Equation (7) bounds the cooling sum of q^2, so a component at least $c\Delta v_1$ survives. Omit cooling if already below threshold. An independent-sign score mixture reduces the tracked component exactly to $v_0 = \Delta e^{-C_3 S}$, without entropy cost.

Holding and prescribed growth. Write $Q = vff^T + R$, with $R \succeq 0$ and diagonal $r \le \Delta$. Leave the perpendicular covariance at its actual value $t^2 R$ and choose covariance $avff^T$ on the scalar target channel. The latent entropy charge is

$$d_t(a) = \tfrac{1}{2}[a/t^2 - 1 - \log(a/t^2)]. \tag{8}$$

With $\psi_s(z) = \mathbb{E}\tanh(z + \sqrt{s}Z)$, the full conditional-mean component is $v'(a) = \mathbb{E}\psi_{t^2 r}(\sqrt{av}Z)^2$. It is continuous and increasing, $v'(1) \le v$, and

$$v'(a) \ge av[1 - F(av + t^2 r)]^2 \ge av[1 - 2(av + t^2 r)].$$

Thus $a \in [1, 1 + C(\Delta + v)]$ can hold v or realize $v_{\text{next}} = \min\{p, v[1 + c_1(\Delta + v)]\}$ exactly as its full conditional-mean component. There is no deliberate component reduction in these steps. Since $\delta \le \Delta/K$, their cost is $O((\Delta + v)^2)$. Conditional Gaussian Poincare gives

$$r' \le t^2 r \mathbb{E}\,\mathrm{sech}^4(\sqrt{av + t^2 r}Z) \le t^2 r(1 - ct^2 r).$$

The last map preserves $[0, \Delta]$ for sufficiently large K. Residuals may increase at small r; only this invariant barrier is used. Holding costs at most $CH\Delta^2$. Prescribed growth has length CS/Δ, cost $C(p + S\Delta)$, and square-root cost $C(S + \log(p/\Delta) + 1)$. These follow by summing the geometric regime $v \le \Delta$ and the increments comparable to v^2 above Δ. Fix the growth sequence before inserting the holding interval.

Purification and scalar continuation. At $v = p \ge \Delta$, holding with $a = 1 + O(p)$ gives $r' \le t^2 r(1 - cp) \le (1 - c'p)r$ if $p \ge K'\delta$. After $C\log(H + 2)/p$ steps, the residual is at most $r_0 = \Delta(H + 2)^{-4}$, at cost $Cp\log(H + 2)$ and square-root cost $C\log(H + 2)$. Take the increasing critical scalar path from p in Imported Theorem 1. For each transition from q to $q' > q$, $\mathbb{E}_\eta G^2/q \ge 1$ because $\tanh^2 G \le G^2$. Consequently

$$\begin{aligned}\mathrm{KL}(\eta\|\mathcal{N}(0, t^2 q)) &= \mathrm{KL}(\eta\|\mathcal{N}(0, q)) + \log t - \tfrac{1}{2}(1 - t^{-2})\mathbb{E}_\eta G^2/q \\ &\le \mathrm{KL}(\eta\|\mathcal{N}(0, q)) + C\delta^2.\end{aligned}$$

Lift each scalar law by $\sqrt{v/q}$ along the tracked target and add the actual perpendicular Gaussian. The residual obeys $r_{j+1} \le t^2 r_j$; over $m \le C/p$ steps it is at most Cr_0, because $\delta/p = O(1)$. The critical heat-convolution estimate preserves the target component up to factor $\exp(-C\sum r_j)$, with extra readout cost $O(mr_0)$. The physical readout cost $1/(8t^2 v_{\mathrm{final}})$ is no larger than its critical comparison. Added suffix entropy is $O(\delta^2/p)$, and added square-root cost is $O(\delta/p)$.

Exact depth and resources. Choose large fixed $B_2 \ge B_1$ and set $\Delta = B_1 U/H$, $p = B_2 U/H$. Taking δ_0 small and the constant in $H \ge CS$ large makes these scales small enough for all estimates. Cooling, growth, purification, and continuation use at most $H-1$ layers; fill the remainder by holding v_0 before growth. Their total action and scalar square-root charge satisfy

$$A \le V_* + C[H^{-1} + H\Delta^2 + S\Delta + p + p\log(H+2) + \delta^2/p + H^{-4}] \le V_* + CU^2/H, \tag{9}$$

$$K_1 := \sum_{j\ge 2} \sqrt{D_j} \le C[H\Delta + S + \log(H+2) + \log(2/p) + \delta/p] \le CU. \tag{10}$$

Every positive tracked component is at least e^{-CU}, and all unit scores are bounded by $B_0 \le e^{CU}$.

Finite width. For the empirical unit score direction u, generate an incoming row as

$$w = N^{-1/2}\{Zu + t(I - uu^T)\xi\}, \qquad \|u\| = 1,$$

where Z has the prescribed scalar law relative to $\mathcal{N}(0,t^2)$ and ξ is independent standard Gaussian. Under that reference, the weight marginal is exactly $\mathcal{N}(0, t^2 I/N)$ conditional on every preceding history. Its entropy is at most the scalar charge. A query comparison has the form $\widehat{c}(x)Z + t\widehat{s}(x)G$ versus its ideal counterpart. Since $\mathbb{E}Z^2 \le t^2 e^{2\sqrt{D_j}}$, the accumulated moment-comparison error is at most $8H\exp(K_1 + H\log t)\sqrt{\alpha}$. The critical bounded-score concentration and stopping proof therefore gives raw probability at least $e^{-N(A+\tau)}$ whenever

$$N \ge CB_0^4 H^4 e^{4K_1 + 4H\log t}(A+1)^6(1+\tau^{-1})^4[\log(2^{d+1}) + \log(H + A + \tau^{-1} + 3)]. \tag{11}$$

The readout noise variance is at most t^2/N, and its shift entropy is divided by t^2, so constants are uniform on the prefactor interval. Explicitly the controlled process has success probability at least $1-\kappa$ and KL at most $N(A + 3A\kappa)$, where $\kappa = \min\{1/4, \tau/[32(A+1)]\}$. Binary entropy yields

$$-N^{-1}\log \mu_{N,t}(E_f) \le \frac{A + 3A\kappa + N^{-1}\log 2}{1-\kappa} \le A + \tau.$$

The width condition enforces $N \ge 8/\tau$. Take $\tau = 1/H$. Every factor in (11) is at most e^{CU}, since $H\log t \le \delta H \le U$. This proves Proposition 1.

Proof of Theorem 1. For $N \ge e^{CU}$, Proposition 1 gives $(B - V_*)_+ \le CU^2/H$; substitution into (5) yields (1), increasing universal constants. For smaller N, the number of real network parameters is at most e^{CU}, and the standard tanh VC bound is polynomial in that count and H. Realizability and exact conditioning make every posterior-supported classifier sample-consistent. Uniform convergence bounds its Gibbs error; the set with correct posterior probability at most 2/3 has mass at most three times that Gibbs error. Choosing the corresponding constant slack proves the claimed inclusive-threshold statement. □

4 Scope and remaining gap

The improved sample dependence is $\exp(O(S + \delta H))$, not $\exp(O(S + \delta^2 H))$. A separate exact Gaussian feature lemma can charge quadratic variance occupation in an equal-diagonal Fock basis, but conditioning on global margin does not preserve Gaussian row laws, and query-dependent variance oracles require additional work. It cannot replace the path-gain argument above without a further theorem. The present positive window is compatible with supercritical lower bounds at longer depth. It neither supplies a lower bound in the canonical range $t \leq 1$ nor proves a polynomial bound there for every bounded-weight Boolean tanh teacher of growing depth.

References

[1] Critical-depth project. Scalar Bellman foundations, *Critical square-root entropy*, *Short near-optimal scalar paths*, and *Finite-width scalar-channel comparison*; source files `critical-square-root-entropy.tex`, `critical-scalar-paths.tex`, and `critical-finite-width-channel.tex`, repository `works-in-progress/critical-depth/source`.

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[3] *Finite-width bounded-dual certificate*, research note `finite-width-dual-certificate.md`, 24 October 2026.

[4] *Critical target mass and the analytic-activation posterior lower bound*, research note `critical-analytic-target-mass.md`, specialized here to exact tanh, 24 October 2026.